AestheticDiagnose — Local AI Assistant for Aesthetic Pre-Consultation Diagnoses

September 2026 · @kaouther

1. Project overview

AestheticDiagnose is a working AI workflow that drafts a first written assessment for cosmetic-surgery patients from their photos and medical file, for the surgeon to review before anything reaches the patient. In a test on 21 real patient files, it processed every file with photos (17 of 17) and produced 13 drafts judged correct (76%), with all inference running locally on a single consumer GPU.

What it is. An n8n automation that watches the clinic's patient database, picks up each file as soon as a surgeon is assigned, retrieves the patient's photos, sends photos and medical context to a vision-language model (Gemma 4 12B, run locally through Ollama), and emails the surgeon a ready-to-review draft written in the surgeon's voice.

Who it is for. Plastic surgeons in a private aesthetic clinic who receive online requests from patients, often international, and must answer each one with a personalised written opinion before a consultation or quote.

Type of solution. A hybrid AI workflow: rule-based automation (database triggers, loops, checks) plus a multimodal AI model plus a human validation step.

Why it matters. Patients wait days for a first answer, and surgeons spend clinical time writing long, similar letters. AestheticDiagnose turns that into minutes of review time per file, without sending patient photos to any external AI service.

How this report follows AI-THINK. Each stage of the method maps to one section:

AI-THINK stage	What it means here	Section
T — Tune	Understand the surgeon's real bottleneck and its root causes	2
H — Hypothesis	Personas, alternative solutions, choice of the best one	3
I — Implement	Build the MVP: architecture, workflow, prompts	4
N — Network	Test on real files, collect feedback	5
K — Knowledge	Decide: end, develop or iterate	7

Responsible AI is treated in its own section (6) because it shaped decisions at every stage, starting with the choice to run the model locally.

2. Problem statement (T — Tune)

Surgeons in a private aesthetic clinic must write a long, personalised first opinion for every online patient request, and this writing, not the medical judgement, is the bottleneck.

Audience. The clinic's plastic surgeons, and indirectly their patients. Patients submit an online file: identity, medical history (diabetes, allergies, past surgery, current treatment), the zones they want treated, up to five requested interventions, free-text wishes, and photos. Once the file is assigned to a surgeon (status 3 in the clinic database), the patient expects a written assessment.

What that letter must contain. A credible first answer is not a template. It must refer to what the surgeon actually sees in the photos, judge whether each intervention is feasible and suitable, explain precautions linked to the medical history, and stay consistent with what the surgeon already told the patient. Patients compare clinics on the quality of this first contact.

Root causes.

- Each letter combines visual analysis, medical cross-checking and careful writing: 4 to 5 rich paragraphs per patient.
- The work is repetitive across similar requests (breast augmentation, liposuction, rhinoplasty) but still needs personalisation.
- Requests arrive continuously; writing competes with consultations and surgery.
- Patient photos and health data are sensitive, so the clinic cannot simply paste them into a public chatbot.

Impact. Slow or generic answers lose patients to competing clinics, and senior surgeon time is spent on drafting rather than on decisions only a surgeon can make.

How AI supported this stage. Problem framing drew on direct work with the clinic's real database and request files, and AI assistants were used to analyse the request structure and summarise the recurring content of surgeon replies.

3. Solution design (H — Hypothesis)

The chosen hypothesis: an automated workflow in which a locally hosted vision-language model drafts the letter and the surgeon stays the only person who can approve it.

Personas

	The surgeon (primary user)	The patient (end beneficiary)
Profile	Plastic surgeon in a private clinic, high caseload, answers online requests between consultations and operations	Adult considering one or several aesthetic procedures, often abroad, comparing several clinics online
Motivations	Convert serious requests into consultations; give answers that reflect their expertise	Know whether the procedure suits them and feel personally understood
Frustrations	Writing long letters for similar cases; delays build up; generic templates sound impersonal	Waiting days for an answer; replies that ignore their photos or their history
Needs	A draft that is already 80–90% right, in their voice, that they can correct in minutes	A fast, clear, honest, personalised first opinion

Alternatives explored

Four directions were tested or seriously considered, and assessed on the three AI-THINK selection criteria.

Option	Impact	Innovation	Feasibility	Outcome
A. Cloud LLM inside n8n (Gemini model node)	High: strong models	Low	High technically, but sends patient photos and health data to an external provider	Used in an early version, then dropped for data protection
B. Local desktop chat tools (Jan, AnythingLLM)	Low: manual copy-paste per patient	Low	High	Useful to test local models, not a workflow
C. Facial geometric measurements + vision model	Medium	High	Medium	Tested: the vision model alone gave better letters than with measurements added
D. n8n workflow + local vision model (Gemma 4 12B via Ollama) + surgeon validation	High: end-to-end, no manual steps until review	High: multimodal AI on private infrastructure	High: runs on an 8 GB consumer GPU	**Selected**

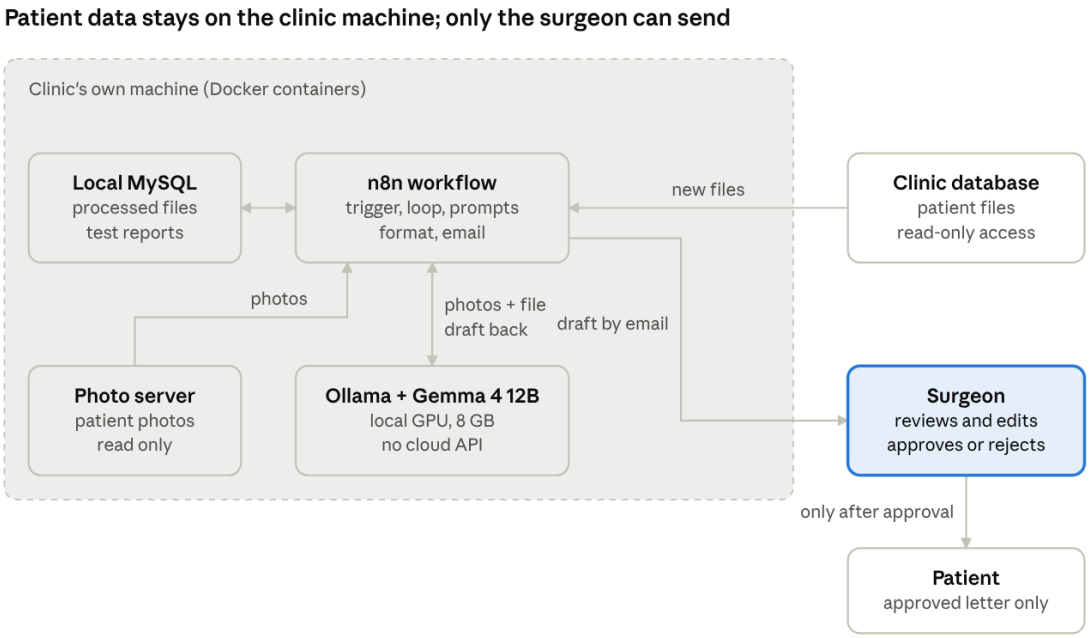
Why option D. It removes the repetitive part of the work (collecting data, looking at photos, drafting) while keeping the two things that must stay human: the medical judgement and the decision to send. It also keeps every photo inside the clinic's own infrastructure.

Model choice. Gemma 4 12B in its QAT (quantisation-aware trained) 4-bit version fits in about 6.3 GB, so it runs on an RTX 5070 laptop GPU with 8 GB of video memory while keeping most of the full model's quality. It reads images natively, which the task requires.

How AI supported this stage. AI assistants were used as a creative partner to compare architectures (cloud versus local, single versus two-step model calls, CPU server plus on-demand GPU server for production) and to draft and refine the prompts.

4. The MVP (I — Implement)

The MVP is a fully working n8n workflow, running on four Docker containers on one Windows PC, that goes from a database change to a draft in the surgeon's inbox with no manual step.



MVP architecture · 4 local services, 1 human approval

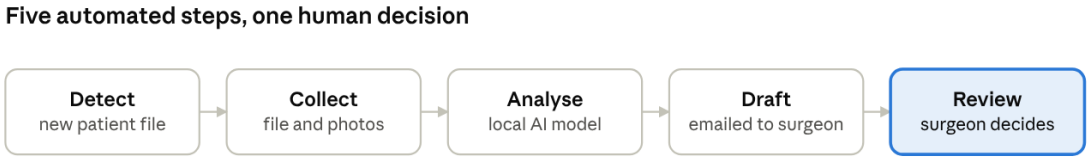
Everything inside the dashed frame runs on one PC; the surgeon is the only path to the patient.

Technical stack

Component	Role	Technology
Orchestration	Trigger, loop over files, data shaping, email	n8n (Docker)
Clinic data	Patient files, interventions, previous surgeon messages (read-only access)	MySQL, clinic patient database
Workflow memory	Tracks which files were already processed; stores test reports	MySQL, separate local database
Photo access	Serves patient photos to the workflow	Lightweight HTTP server on the local network
AI model	Reads the photos and the file, writes the draft	Gemma 4 12B QAT, served by Ollama (Docker)
Hardware	Local inference	RTX 5070 laptop GPU, 8 GB VRAM, 32 GB RAM, WSL2 Ubuntu
Human step	Review, correct, approve	Surgeon's email (Gmail)

How the workflow runs

1. **Detect.** At regular intervals, the workflow checks the clinic database for patient files newly assigned to a surgeon, and keeps its own record so each file is handled only once.
2. **Collect.** For each new file, it gathers the medical history, the requested interventions, the patient's wishes, the surgeon's previous message and the photos. Files without photos are skipped and logged.
3. **Analyse.** The data and photos are sent, with a structured prompt, to the vision model running locally.
4. **Draft.** The model's output is formatted and emailed to the surgeon as a draft to review.
5. **Review and log.** The surgeon decides what is sent. Each result is logged for quality tracking.



Workflow overview · simplified

Figure: simplified view of the workflow. The full n8n implementation is available on request.

Prompt engineering

The prompts went through two versions.

- **Version 1** asked for a long, multi-paragraph assessment based on the photos, the history and the surgeon's previous exchange, written as the surgeon.
- **Version 2** added an explicit hidden reasoning step before writing: map visual landmarks in the photos, check anatomical feasibility, assess risks from the medical history, then decide the overall aesthetic goal. It also fixed the format (4–5 paragraphs, no lists, first person) and a "safety first" rule: if diabetes, an allergy or a previous surgery makes a procedure risky, that must be addressed.

The user prompt is filled automatically from the database and only includes the interventions actually requested (up to five).

Production architecture (designed, not yet deployed)

For real use, the design splits the system in two: an always-on, low-cost CPU server (n8n, database, photo storage) and a GPU server started only when there are files to analyse, connected through a private encrypted tunnel (WireGuard or Tailscale). The GPU is the expensive part, so it should not run idle.

> *Screenshots to add here: (1) one example email received by the surgeon, fully anonymised; (2) the test-report sheet with names removed.*

5. Validation and feedback (N — Network)

On 21 real patient files, AestheticDiagnose processed all 17 files with photos and produced 13 correct drafts (76%); the 4 failures fall into two clear, fixable categories.

Test set-up

The MVP was tested in a controlled, private environment, the testing scope AI-THINK recommends before any wider exposure: real historical files from the clinic, processed end to end by the workflow, with outputs reviewed one by one instead of being sent to patients. Each draft was read and classified as correct or not, with the reason noted, in the automatically generated Excel test report.

Measure	Result
Files processed by the workflow	21
Files without photos (correctly detected and skipped)	4
Files with photos, draft generated	17 of 17
Drafts judged correct	13 of 17 (76%)
Procedures covered	Breast augmentation and lift, abdominoplasty, liposuction, rhinoplasty, face and neck lift, blepharoplasty, fat-transfer procedures

What went wrong

Failure	Cases	What happened	Likely cause
Truncated letter	2 (both rhinoplasty)	The letter stops mid-sentence	Output length limit or formatting after the model call (under investigation)
Unsupported visual assumption	1	The letter assumed the patient was lying down in the photo	The model describes beyond what it can see
Factual error on age	1	The letter stated a wrong age	The model inferred instead of using the date of birth from the file

One more observation: a patient asked for a buttock augmentation in the free-text "other requests" field, and the draft focused on the structured interventions. Whether "other requests" should carry equal weight is a design decision for the clinic.

What worked well: every correct letter used observations from the photos, followed the 4–5 paragraph format, addressed the patient in the surgeon's voice, and integrated the medical history.

Feedback received

Programme instructor (on the architecture). The feedback confirmed the strongest design choice, keeping the surgeon as the final decision-maker, and praised the "smallest system that respects the data" approach and the move to local deployment. It asked for four improvements: no direct sending to patients even after approval without traceability and a controlled template; documenting failure cases of the vision model, not only good outputs; a risk matrix (data, clinical, communication, operational) with audit logs, consent and retention rules; and an evaluation rubric (accuracy, expression of uncertainty, refusal behaviour, avoidance of medical decisions).

Surgeons. Structured feedback from the surgeons has not been collected yet. It is the next validation step (section 7), with a short form after each reviewed draft: usable as is / minor edits / major edits / unusable, time spent on review, and one comment.

How AI supported this stage. The workflow itself generated the test report, and AI assistants helped classify failure patterns and turn the instructor's feedback into concrete changes (the checker loop described in section 7).

6. Responsible AI considerations

AestheticDiagnose handles faces, bodies and health data, so responsibility was designed in from the start: the model runs locally, it only drafts, and a surgeon decides.

Principles applied

- **Human in the loop, always.** The AI never contacts a patient. It sends a draft to the surgeon, who corrects, approves or discards it. The surgeon remains medically and legally responsible for every word sent.
- **Privacy by design.** Photos and medical data never leave the clinic's own machine: no cloud AI API is called. Access to the clinic database is read-only, and the workflow's own records are kept in a separate local database.
- **Data minimisation.** Only the fields needed for the letter are extracted. Test material shown in this report and the demo video is anonymised (no patient names, faces cropped or replaced, no surgeon names).
- **Honest reporting of failures.** The 24% of incorrect drafts are documented with their causes, not hidden.

Risk matrix

Risk area	Risk	Current mitigation	Next step
Data	Leak of patient photos or health data	Local inference; read-only database access	Restrict the photo server to the internal Docker network; encrypt the GPU-server link in production;

Risk area	Risk	Current mitigation	Next step
			written retention rules
Clinical	The model invents an observation or misses a contraindication	Surgeon review of every draft; "safety first" prompt rule	Automatic checker for known failure types; rubric scoring
Communication	the patient misled about who wrote the letter	Surgeon approval before sending	decide how AI assistance is disclosed to patients
Operational	Truncated or empty output reaches the surgeon unnoticed	Test-report logging	Checker loop with retry and error alert
Fairness	Visual analysis less accurate for some skin tones, ages or body types	None specific yet	Track accuracy by patient group as the test set grows

An honest tension: transparency

An open question is disclosure: the letter is written in the surgeon's voice. Because the surgeon reviews and signs it, it is the surgeon's letter, but the clinic should decide, with legal advice, whether and how patients are told that an AI assistant prepared the first draft.

7. Key learnings and conclusion (K — Knowledge)

The decision is to **iterate, then develop**: the MVP proves the technical concept, the failures are understood and fixable, but surgeon feedback is still needed before scaling.

Key learnings

1. **A small local model is enough for a real professional task.** A 12B model on an 8 GB laptop GPU wrote usable surgeon letters from real photos 3 times out of 4. Privacy did not have to be traded for capability.
2. **Simpler can be better.** Adding geometric facial measurements did not improve the letters; the vision model reading the photos directly did better.
3. **Generating is not the hard part; checking is.** Every failure (cut-off letters, invented details, wrong age) would have passed a quick glance. A workflow needs an explicit checker before it can be trusted, the principle "no checker, no unattended agent."
4. **Engineering details decide reliability.** Hard-coded IP addresses, container networks and data volumes caused more lost time than the AI model. Treating infrastructure as code (one Docker Compose file) is part of building responsibly.
5. **The human role changes, it does not disappear.** The surgeon moves from writing to reviewing and deciding, which is where their expertise matters most.

Next steps

Step	Goal	Success measure
Add a checker node with retry	Catch truncated letters, wrong age, unsupported visual claims; retry up to 3 times, then raise an alert	Correct drafts from 76% towards the 95% target
Prompt version 3	Keep the warm tone, state risks clearly, use the date of birth for age, describe only what is visible	No missed contraindication in the test set
Surgeon pilot	Each surgeon rates every draft and notes review time	At least 20 rated drafts; median review time
Harden the infrastructure	One Docker Compose file for all services; photo server reachable only inside Docker	Restart without manual reconfiguration
Production design	CPU server always on, GPU server on demand, private tunnel	Cost per analysed file
Governance	Consent wording, retention rules, impact assessment, AI-assistance disclosure	Validated by the clinic before go-live

Conclusion

AestheticDiagnose shows how AI-THINK turns a concrete professional bottleneck into a working, responsible AI solution. Tuning the problem showed that the burden was the writing, not the medical decision. The hypothesis stage led to a local, human-validated design instead of the easier cloud option. The MVP runs end to end on real data, and the tests showed both what it can do and exactly where it fails. The next cycle will add the checker, the surgeons' own feedback and the governance needed before a patient ever receives a letter that started as an AI draft.

Appendix A. Evidence checklist

- ☐ One anonymised draft email as received by the surgeon
- ☐ Excel test report with names removed (section 5)
- ☐ Terminal showing `docker ps` with n8n, MySQL and Ollama running, and `ollama list` showing Gemma 4

Appendix A. Evidence checklist

- Anonymized Email :



Hi Dr

Your partial diagnosis for client 116 is now ready.

Please log in to the app to review the details.

Thank you,

The . Team

Partial diagnosis :

Je suis le Dr ., chirurgien plasticien spécialisé dans la chirurgie esthétique. C'est avec une attention particulière que j'ai pris connaissance de votre dossier et des objectifs que vous souhaitez atteindre afin d'harmoniser votre silhouette. Votre profil clinique est excellent, avec un indice de masse corporelle sain et une absence d'antécédents médicaux notables tels que le diabète ou les allergies, ce qui constitue une base favorable pour la réussite de vos interventions. À travers l'examen attentif des photographies fournies, j'ai pu identifier les zones clés de votre demande : une ptose mammaire significative accompagnée d'un volume important, ainsi qu'une distribution adipeuse marquée au niveau du bas de l'abdomen et des faces antéro-latérales des cuisses. Mon approche consiste à créer une harmonie globale en traitant ces volumes avec précision, tout en garantissant un résultat naturel qui respecte votre morphologie.

Pour atteindre vos objectifs esthétiques, nous devons adopter une stratégie cohérente qui dépasse la simple exécution technique de chaque intervention isolée. Concernant le projet mammaire, bien que vous ayez exprimé le souhait d'une augmentation par prothèses rondes, l'évaluation visuelle révèle une ptose importante et un étirement vertical du tissu cutané et glandulaire. Pour obtenir un résultat élégant et durable, il est essentiel de prendre en compte le remodelage des tissus dès la planification chirurgicale ; une simple augmentation pourrait ne pas suffire à corriger l'affaissement actuel. De même, pour le remodelage corporel, la liposuccion est parfaitement indiquée pour affiner les flancs ainsi que les régions quadriceps et trochantériennes des cuisses. Toutefois, au vu de la présence d'un pannicule abdominal inférieur associé à un relâchement cutané, il conviendra de discuter si une approche complémentaire est nécessaire, car la liposuccion seule n'est pas toujours suffisante pour traiter l'excès de peau dans cette zone spécifique afin d'obtenir un profil parfaitement lisse.

La sécurité de votre parcours chirurgical est ma priorité absolue, et les protocoles d'anesthésie générale associés à ces interventions sont conçus pour assurer votre confort et votre sérénité durant toute la durée des opérations. Nous porterons une attention particulière à l'élasticité de vos tissus et à leur capacité de cicatrisation pour garantir une récupération optimale. Je tiens à préciser que cette consultation est préliminaire et repose sur les éléments visuels transmis ; elle définit les orientations majeures de votre projet esthétique sans en constituer le plan définitif. Si vous décidez de poursuivre notre collaboration, le plan chirurgical détaillé sera confirmé lors de l'examen clinique complet en présentiel, étape indispensable pour valider la faisabilité technique et personnaliser chaque geste. Mon équipe de coordination patient se tient d'ores et déjà à votre entière disposition pour vous accompagner avec bienveillance dans les prochaines étapes de votre parcours vers une silhouette sublimée.

This email was sent automatically with n8n

Répondre Transférer

- Excel test report

id	id_client	id_devis	status	created_at	interventions demandées	Reviewer Analysis
48	1	1	processed	2026-07-29 15:22:17	Liposuccion 3 zones (zones posterieures non incluses)	ok
49	9	8	processed	2026-07-29 15:27:53	Abdominoplastie + Augmentation mammaire -prothèses rondes-	ok
50	12	10	processed	2026-07-29 16:34:13	lifting seins avec prothèses rondes + Abdominoplastie	ok
51	49	29	processed	2026-07-29 15:40:26	Rhinoplastie complete	diagnotic tronqué!
52	58	35	processed	2026-07-29 15:46:48	lifting seins avec prothèses rondes	ok
53	63	42	processed	2026-07-29 15:52:56	lifting seins sans prothèses	ok
54	66	44	processed	2026-07-29 16:58:04	brest uplift with implants- round implants	ok
55	68	45	processed	2026-07-29 16:03:10	liposuccion deux zones, zones posterieures non incluses	ok
56	69	46	processed	2026-07-29 16:08:08	Rhinoplastie complete	ok
57	74	51	processed	2026-07-29 17:13:09	liposuccion deux zones, zones posterieures non incluses	position allongée supposée
58	82	59	processed	2026-07-29 16:18:28	penoplastie par injection de graisse	erreur âge
59	86	63	processed	2026-07-29 17:23:46	augmentation mammaire prothese ronde + lifting seins prothese rondes	ok
60	89	66	processed	2026-07-29 16:28:50	liposuccion deux zones, zones posterieures non incluses + penoplastie par injection de grai	ok
61	97	74	processed	2026-07-29 17:34:14	lifting cervico facial complet + blepharoplastie)	ok
62	99	76	processed	2026-07-29 16:39:33	Rhinoplastie complete	tronqué !
63	103	80	processed	2026-07-29 16:45:26	augmentation mammaire prothese ronde (+ augmentation de fesse mais mise dans autres,	ok
64	107	84	Skipped	2026-07-29 16:45:27		
65	114	91	Skipped	2026-07-29 16:45:27		
66	116	93	processed	2026-07-29 17:50:24	augmentation mammaire prothese ronde + liposuccion trois zones, zones posterieures no	ok
67	118	95	Skipped	2026-07-29 16:50:24		
68	121	98	Skipped	2026-07-29 16:50:25		
				21		
				17		
				13		
				76,47%		

- Terminal showing `docker ps`

```
[redacted]@DESKTOP-9TH01DP:~/project[redacted]$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED       STATUS       PORTS
                     NAMES
d1701c95544c   n8nio/n8n:latest   "tini -- /docker-ent…"   4 hours ago   Up 4 hours   0.0.0.0:5678->5678/tcp, [::]:5678-
>5678/tcp            n8n
65522f40d671   mysql:8.0          "docker-entrypoint.s…"   4 hours ago   Up 4 hours   0.0.0.0:3306->3306/tcp, [::]:3306-
>3306/tcp, 33060/tcp   mysql
c35ef7d6f3e0   ollama/ollama      "/bin/ollama serve"      4 hours ago   Up 4 hours   0.0.0.0:11434->11434/tcp, [::]:114
34->11434/tcp          ollama
```